Modelling the Efficiency of a Sheep Pasture in Minecraft

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Abstract

valarante child game.... look to cartoon grapfix to make kid player happy like children show.. valarante cartoon world with rainbow unlike counter strike with dark corridorr and raelistic gun.. valarante like playhouse. valarant playor run from csgo fear of dark world and realism

1 Introduction

Minecraft is an open-world sandbox video game[1]. The game takes place in a procedurally generated world made of cubic voxels, or *blocks*. Among other activities, players can engage in agriculture, including the raising of sheep. Sheep can be sheared sheared to gather wool, and require grass to eat to regrow their wool after shearing. Grass blocks are converted to dirt after being eaten, and may be converted back to grass by other adjacent grass blocks. If a pasture is packed too densely with sheep, they will deplete the grass, and wool production will be low. However, if the pasture is not densely packed, wool production will still be low because there are not many sheep. Between these two extremes, there is an optimal sheep population that will maximize wool output.

1.1 Deterministic vs. Stochastic

Shortly, we will seek a model for the population of grass in a pasture. The growth and consumption of grass is a stochastic process, like everything else in *Minecraft*, and also like population evolution in the real world. Early population models were deterministic, but more sophisticated and useful models are stochastic, incorporating the uncertainty inherent in population size. This paper will take the deterministic approach, owing to its simplicity[2].

1.2 A brief tangent about units

Canonically, one block is one cubic meter.

[1]This investigation was conducted in Minecraft version 1.19.4.
[2]And the author's modest mathematical background.

2 Key assumptions and notes

There are a couple key assumptions being made here:

- Grass growth is a simple process in which all dirt blocks have an equal chance to turn into a grass block on a tick. This is not entirely realistic, since grass growth is contingent on spreading from an adjacent block, but as long as dirt blocks consistently have adjacent grass blocks it should work.
- The actual rate at which the farm produces wool is limited by the rate at which the farmer shears the sheep. Initially, we will assume the farmer is infinitely fast, so really the goal is to have the greatest rate of sheep eating grass. Later, we will look at the mechanics of a farm where the farmer harvests the wool at regular intervals.
- Another important thing to note is that pastures can be discussed in terms of *blocks* or *area*. One block is canonically one square meter, So pasture size has units of area. However, area in minecraft is discretized, i.e. you cannot have fractional area. For this reason, pasture area will be discussed using the unitless quantity *blocks*.

3 Model

3.1 Grass consumption by sheep

Sheep attempt to eat at a rate R [s^{-1}]. A sheep's attempt to eat will only succeed if it is standing on a grass block. A pasture containing N sheep, whose surface is covered in grass with fraction X_g, will lose grass at a rate

$$NRX_g \tag{1}$$

as it is eaten by sheep. This assumes that the sheep are distributed uniformly so that the probability of any sheep standing on a grass block is equal to the grass cover of the pasture.

3.2 Grass growth

The rate at which grass grows not as simple. If X_g is small, then grass growth will be limited by the lack of places the grass can spread *from*. If X_g is close to 1, grass growth will be limited by a lack of places for the grass to spread *to*. We will employ a logistic model, where population follows the differential equation:

$$\dot{y} = ky\left(1 - \frac{y}{p}\right)$$

Where $y(t)$ is the population at time t, k is a parameter representing the reproduction rate of the population and p is the carrying capacity, the maximum population that the local environment can sustain.

The logistic model is one of the oldest models for population growth. It was born from the older exponential (or *Malthusian*) model, but modified to include a maximum population size. It is useful as a tool of instruction, but it's not very useful for modelling actual populations, and has since been supplanted by more sophisticated models. However, we are not dealing with an actual population, but rather a video game which has simple population mechanics. Therefore the model is appropriate for our purposes. Defining A as the total number of blocks in the pasture, we will consider the grass population AX_g, with carrying capacity A. The equation becomes the following:

$$A\dot{X}_g = kAX_g(1 - X_g) \tag{2}$$

Where k [s^{-1}] is a constant as described above. The carrying capacity is 1 because naturally, the pasture cannot be more than 100% covered in grass.

3.3 The fast farmer

If the pasture reaches equilibrium, the rate at which the sheep eat should be equal to the rate at which the grass is regrowing. We can express this, by combining (1) and (2), as

$$NRX_g = A\dot{X}_g = kAX_g(1 - X_g)$$

This expression is important because it gives us the equilibrium grass cover as a function of sheep population - a little algebra gives

$$X_g = 1 - \frac{NR}{Ak} \tag{3}$$

Plugging this into our logistic model for grass growth (2) gives

$$A\dot{X}_g = NR - \frac{N^2R^2}{Ak}$$

This is the steady-state grass regrowth rate as a function of sheep population. If the farmer is shearing sheep as quickly as they get their wool, they are getting 1 block of wool for every block of grass the sheep eat. Let's define F [s^{-1}] as the rate at which the farmer gathers wool.

$$F_{fast} = A\dot{X}_g = NR - \frac{N^2R^2}{Ak} \tag{4}$$

We can find the maximum by differentiating (4) with respect to N, setting the derivative to 0 and solving for N:

$$\frac{dF_{fast}}{dN} = R - \frac{2NR^2}{Ak} = 0$$

$$N = \frac{Ak}{2R} \tag{5}$$

According to the assumptions we've made, this is the sheep population that will produce the maximum rate of grass growth. At this population,

$$F_{fast} = \frac{Ak}{4}$$

3.4 The realistically slow farmer

For the case of the infinitely fast farmer, the rate at which wool is harvested is the same as the rate at which the sheep are eating. To account for the farmer being slow, we'll need to think a bit more.

The farmer is off doing other things, and every T seconds, they come back to shear the sheep. Upon shearing the sheep, all the sheep are wool-less. By the time they come back for the next shearing, the sheep have regrown some amount of wool. Let's say the number of sheep that have their wool is N_w. So when the farmer shears the sheep, they get N_w wool. Then the average rate at which the farmer gathers wool is

$$F_{slow} = \frac{N_w}{T}$$

Now we need to consider the value of N_w. A sheep will only regrow wool upon eating if it doesn't already have any, so we should expect that the rate at which sheep are regrowing wool is the rate at which they are eating, NRX_g, times the fraction of sheep that are wool-less:

$$\dot{N}_w = \frac{dN_w}{dt} = \frac{N - N_w}{N} NRX_g = (N - N_w)RX_g$$

At time $t = 0$, the farmer shears all the sheep, leaving them with $N_w = 0$, and then at time $t = T$, the farmer comes back and shears them again, gathering N_w blocks of wool. How many sheep grew their wool back in time T [s]? We can find out by solving the above equation for N_w. We'll separate it into N_w terms and non N_w terms:

$$\frac{1}{N - N_w} dN_w = (RX_g)dt$$

and integrate from initial state $(t = 0, N_w = 0)$ to the final state $(t = T, N_w = N_w(T))$. We're assuming the pasture has reached steady state, so X_g is independent of t and can be treated as a constant.

$$\int_0^{N_w(T)} \frac{1}{N - N_w} dw = \int_0^T RX_g dt$$

$$-\ln\left(\frac{N - N_w(T)}{N}\right) = TRX_g$$

And solve for $N_w(T)$:

$$N_w(T) = N\left(1 - e^{-TRX_g}\right)$$

So when the farmer comes back to shear the sheep at $t = T$, they will get $N - Ne^{-TRX_g}$ blocks of wool. If the farmer keeps coming back every T seconds, they will be gathering wool at a rate

$$F_{slow} = \frac{N\left(1 - e^{-TRX_g}\right)}{T}$$

Finally, we can plug (3) for X_g:

$$F_{slow} = \frac{N\left(1 - e^{-TR(1-\frac{RN}{Ak})}\right)}{T} \tag{6}$$

This is the expression for the rate of wool gathering by a slow farmer. The optimal density of sheep is now dependent on T. To find the optimal sheep population, we can take the same approach as we did with the fast farmer case. Differentiating (6) w.r.t. N:

$$\frac{dF}{dN} = \frac{1 - \left(\frac{TR^2N}{Ak} + 1\right)e^{-TR+\frac{TR^2N}{Ak}}}{T} = 0$$

Attempting to solve for N leads to

$$\left(\frac{TR^2N}{Ak} + 1\right)e^{\frac{TR^2N}{Ak}+1} = e^{TR+1}$$

At this point we must invoke the lambert W function, a nonelementary function satisfying

$$W(x)e^{W(x)} = x$$

applying this leads to

$$W(e^{TR+1}) = \frac{TR^2N}{Ak} + 1$$

$$N = \frac{Ak}{TR^2}(W(e^{TR+1}) - 1) \tag{7}$$

Plugging this back into (6), we get

$$F_{slow} = \frac{Ak(W(e^{TR+1}) - 1)(1 - e^{-TR+W(e^{TR+1})-1})}{R^2T^2}$$

4 Experimental determination of rate constants

Before proceeding further, it's worth determining the actual values of k and R. This will be done via in-game experiments, which will be described briefly here. More detail can be found on the author's github[3].

[3]github.com/ELFREELAND/minecraft_sheep

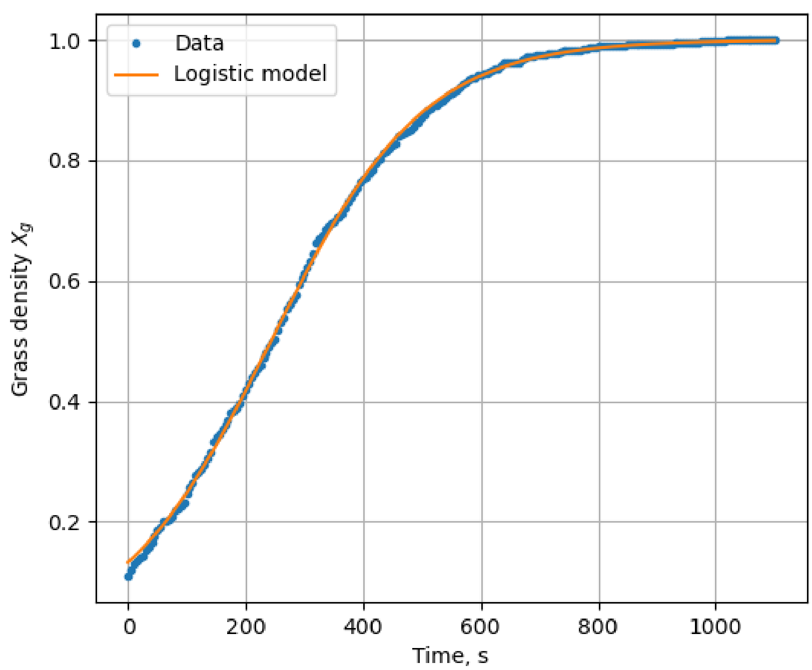
Ten sheep were locked in cells with a grass block immediately underneath them. The cells included circuitry to count when each sheep ate the grass and automatically replenish it. In one hour, the sheep consumed 320 blocks of grass. This gives an R value of 0.00888 s^{-1}.

A 30x30 patch of dirt was isolated from its surroundings and uniformly interspersed with 100 grass blocks. As the grass grew, the grass population was recorded every 5 seconds. The data was fit to the exact solution of the logistic equation:

$$y = \frac{1}{1 + e^{-k(x-x_0)}}$$

The best fit gave $k = 0.00772$. The data and fit are compared in Figure 1. The close fit gives us also confidence that the logistic model is appropriate.

Figure 1: Logistic model fit to grass growth data



5 Model Discussion

Figure 2 shows F_{fast} and X_g as a function of N, normalized for pasture area. These are simple curves,a parabola and straight line respectively, but there are a few important features to glean. First, the equilibrium X_g and F_{fast} are both zero at $\frac{N}{A} = \frac{k}{R} = 0.869$. Deterministically, a pasture with this density will lose all of its grass.

Second, the maximum F_{fast} occurs at $X_g = 0.5$. This is a feature of the logistic model; population growth is greatest when the population is half of the carrying capacity.

Figure 2: Plot of $\frac{F_{fast}}{A}$ and X_g as a function of N.

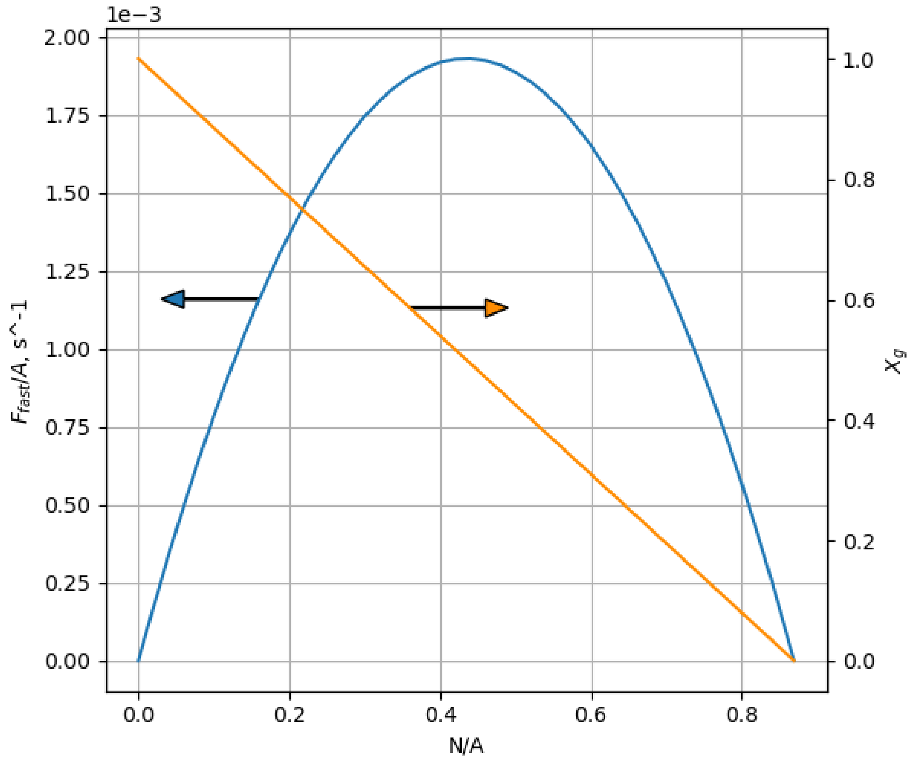


Figure 3 shows F_{slow} as a function of N and T. For the case of $T = 1$, F_{slow} follows the same parabolic pattern with respect to N as F_{fast}. We can see this algebraically by employing the following approximation, for small x:

$$e^x - 1 \approx x$$

Applying this to (6):

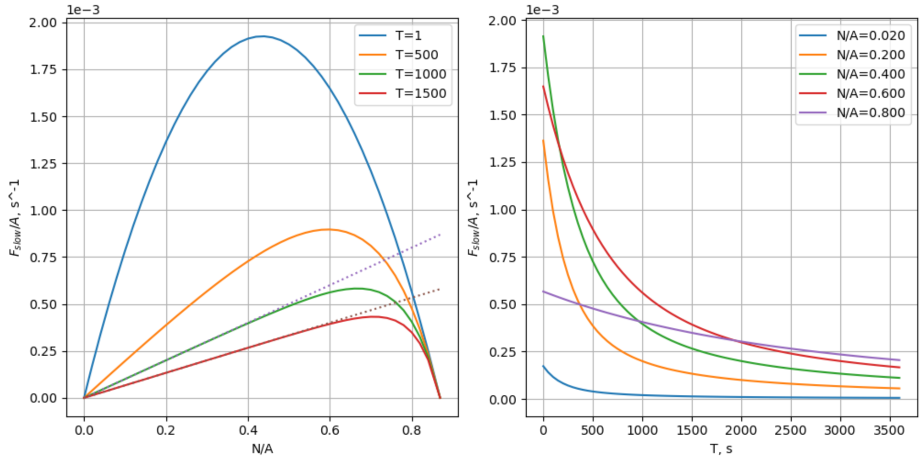
$$F_{slow} \approx NR - \frac{N^2R^2}{Ak} = F_{fast}$$

For large values of T and modest values of N, F_{slow} is approximately linear w.r.t. N. Algebraically, the term $e^{-TR(1-\frac{RN}{Ak})} \approx 0$ in (6) for large negative values of the exponent. Under these conditions,

$$F_{slow} \approx \frac{N}{T}$$

Lines of $\frac{N}{T}$ are plotted over top of the curves in figure 3.

Figure 3: Plots of $\frac{F_{slow}}{A}$ as a function of N and T.



6 Stochastic model

Thus far we have considered a deterministic model for the grass population of the pasture. This allows for convenient exact solutions for the population evolution, but

$$\Pr(AX_g \uparrow x, t = t_0 + \Delta t) = Poisson(A\dot{X}_g \Delta t) = \frac{(A\dot{X}_g \Delta t)^x e^{-A\dot{X}_g \Delta t}}{x!}$$

$$\Pr(AX_g \downarrow x, t = t_0 + \Delta t) = Poisson(NRX_g \Delta t) = \frac{(NRX_g \Delta t)^x e^{-NRX_g \Delta t}}{x!}$$